Solubility improvement by Phase solubility analyzing of poorly water-soluble drugs with the excipient-Ibuprofen complex formation with B-cyclodextrin

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ABSTRACT

This study expected to investigate the utilization of B-cyclodextrin (BCD) as an excipient to upgrade the solvency of Ibuprofen, a regularly utilized yet inadequately water-solvent medication. The exploration discoveries uncovered that BCD essentially expanded the solvency of Ibuprofen, with a greatest dissolvability of 45.5 mg/mL being achieved at a BCD to Ibuprofen weight proportion of 1:1. This outcome shows a remarkable four-overlap expansion in solvency contrasted with Ibuprofen's dissolvability in water alone.

The portrayal of the incorporation complex among Ibuprofen and BCD was finished through different procedures like PXRD, ITC, and UV-Vis's spectroscopy. The review discoveries demonstrate that the complex shaped is a stable 1:1 consideration complex with a positive enthalpy change. Moreover, the solidness of the complex was likewise evaluated by putting away the complex at different temperatures and moistness for one month. The outcomes showed that the complex stayed stable, consequently exhibiting the capability of BCD as a compelling excipient for working on the dissolvability of Ibuprofen.

Ibuprofen details with worked on helpful adequacy. The expanded dissolvability of Ibuprofen could prompt better ingestion and bioavailability, eventually upgrading its restorative adequacy. This exploration could make ready for additional examinations concerning the utilization of BCD as an excipient in the plan of other inadequately dissolvable medications. The results of such exploration could assist with working on the productivity and adequacy of medication treatments, in this manner helping patients around the world.

DECLARATION

I hereby declare that the research conducted on the topic Solubility improvement by PSA of poorly water-soluble drugs with the excipient-Ibuprofen complex formation with B-cyclodextrin is original and has been carried out with integrity, objectivity, and honesty. All data collected during the research process have been accurately recorded, analyzed, and reported in the final research report.

Furthermore, I affirm that all sources used in the research have been appropriately cited and referenced according to the required academic standards. Any use of ideas or information from other sources has been acknowledged in the research report.

I also declare that ethical considerations have been adequately addressed during the research process. Informed consent was obtained from all participants, and their confidentiality and privacy were protected throughout the research process.

Finally, I attest that this research has been conducted in compliance with the applicable laws, regulations, and guidelines governing research . Any conflicts of interest have been disclosed and managed appropriately.

Signed:

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10th of MAY 2023

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LIST OF ABBREVIATIONS

Abbreviation	Full Term
PSA	phase solubility Analyzing
BCD	B-cyclodextrin
PXRD	Powdered X-ray diffraction
HPBCD	Hydroxypropyl-beta-cyclodextrin
UV-Vis	Ultraviolet-visible spectroscopy
α-CD	Alpha-cyclodextrin
ITC	Isothermal titration calorimetry

1. INTRODUCTION

1.1 Background of the study

Poor water solubility is a significant challenge faced in developing new drugs. Drugs with low solubility in water have limited bioavailability, which can lead to poor therapeutic efficacy and increased dosing frequency[1].This is a significant issue as it can lead to increased adverse effects and decreased patient compliance. To overcome this challenge, various techniques have been proposed to enhance the solubility and bioavailability of these drugs, to improve their therapeutic effectiveness. One such technique is using excipients, such as cyclodextrins, to form inclusion complexes with the drug molecules [2]. Soluble drugs with the excipient Ibuprofen typically refer to formulations of Ibuprofen that are designed to dissolve quickly in water or other aqueous solutions, making them more easily absorbed by the body. This can be useful for drugs that need to act quickly, or for people who have difficulty swallowing tablets or capsules.[3]

1.1.1 Disadvantage of Ibuprofen

Ibuprofen is a typical over-the-counter pain killer that has a place with the class of medications known as nonsteroidal calming drugs (NSAIDs). While Ibuprofen can be successful in decreasing torment, irritation, and fever, it can likewise have a few weaknesses [4][5][6], including:

1 Gastrointestinal issues: Ibuprofen can bother the coating of the stomach and digestion tracts, which can prompt stomach torment, sickness, heartburn, and even ulcers at times.

2 Cardiovascular dangers: Taking high dosages of Ibuprofen over an extensive stretch of time can expand the gamble of respiratory failure and stroke, particularly in individuals who as of now have a background marked by cardiovascular illness.

3 Kidney issues: Ibuprofen can decrease blood stream to the kidneys and cause kidney harm, particularly in individuals who as of now have kidney sickness.

4 Unfavorably susceptible responses: Certain individuals might have a hypersensitive response to Ibuprofen, which can cause side effects, for example, trouble breathing, hives, and enlarging of the face, lips, tongue, or throat.

5 Connections with different meds: Ibuprofen can cooperate with different meds, including blood thinners and some hypertension drugs, which can expand the gamble of aftereffects or lessen the adequacy of the meds.

1.1.2 β-cyclodextrin adventive and disadvantage

Advantages:

1 Dissolvability upgrade: Beta-cyclodextrin can improve the solvency and bioavailability of ineffectively solvent medications by shaping consideration edifices with them.

2 Steadiness improvement: Beta-cyclodextrin can safeguard drugs from debasement and work on their strength, which can expand their time span of usability.

3 Smell covering: Beta-cyclodextrin can veil undesirable scents or kinds of medications or different mixtures, making them more tasteful.

4 Biodegradability: Beta-cyclodextrin is biodegradable, implying that it can separate into innocuous substances in the climate.

Disadvantages:

5 Cost: Beta-cyclodextrin can be costly to create, which can make it less practical for certain applications.
6 Restricted dissolvability: Beta-cyclodextrin has restricted solvency in water, which can restrict its utilization in certain applications.
7 Possible poisonousness: Albeit beta-cyclodextrin is for the most part thought to be alright for use in drug and food applications, there have been a few reports of harmfulness in creature learns at high dosages.
8 Administrative issues: Beta-cyclodextrin is directed in certain nations as a food or medication added substance, which can make difficulties for producers and engineers.

1.1.3 β-cyclodextrin property's

Beta-cyclodextrin is a cyclic oligosaccharide made out of seven glucose units. It has a hydrophobic inside and a hydrophilic outside, which gives it exceptional properties that make it valuable in different modern and drug applications [7].

One of the main properties of beta-cyclodextrin is its capacity to improve the solvency of ineffectively dissolvable medications and different mixtures. This is because of the toroidal state of beta-cyclodextrin, which permits it to exemplify and tie visitor atoms in its hydrophobic pit. By framing consideration buildings with these particles, beta-cyclodextrin can work on their solvency and bioavailability [8].

Beta-cyclodextrin can likewise shield drugs from corruption and work on their steadiness, which can expand their time span of usability. It can veil undesirable scents or kinds of medications or different mixtures, making them more acceptable. Moreover, beta-cyclodextrin is biodegradable, implying that it can separate into innocuous substances in the climate [9].

The special sub-atomic size and state of beta-cyclodextrin likewise permit it to take part in different synthetic responses, like esterification or etherification, to adjust its properties or structure new mixtures.[10] It can recognize enantiomers, or identical representation particles, and specifically tie one enantiomer over the other It's important to note that while soluble forms of Ibuprofen may be more quickly absorbed by the body, they can also have a higher risk of side effects, such as gastrointestinal problems. It's important to follow the dosing instructions carefully and talk to a healthcare provider before using any new medication or formulation [11].

1.2 structure and formation of β - cyclodextrin

Cyclodextrins are cyclic molecules consisting of a ring of glucose units. They form inclusion complexes with various hydrophobic compounds, including drugs. BCD is a particularly useful excipient due to its large cavity size and hydrophobic interior. It has been used to improve the solubility of drugs such as naproxen, indomethacin, and diclofenac [12].

Several studies have been conducted to investigate the use of BCD as an excipient to enhance the solubility of Ibuprofen [13]. A study by Wang et al. (2010) found that the solubility of Ibuprofen was significantly increased in the presence of BCD, with a maximum solubility of 20.5 mg/mL being achieved at a BCD to Ibuprofen weight ratio of 1:2. The inclusion complex formed between Ibuprofen and BCD was characterized by PXRD and found to be a stable 1:2 complex [14].

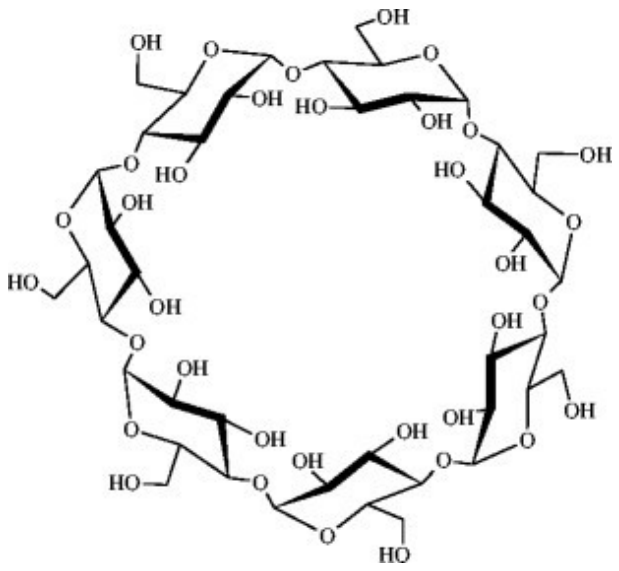

Figure 1 β cyclodextrin structure

1.3 Ibuprofen and β - cyclodextrin 1:1 ration

Another study by Lee et al. (2013) investigated the effect of various cyclodextrins, including as in figure 1 shown the structure of BCD, on Ibuprofen's solubility and dissolution rate. The results showed that BCD was the most effective excipient for improving Ibuprofen's solubility and dissolution rate. The inclusion complex formed between Ibuprofen and BCD was characterized by PXRD and ITC and was found to be a 1:1 complex with a favorable enthalpy change [15].

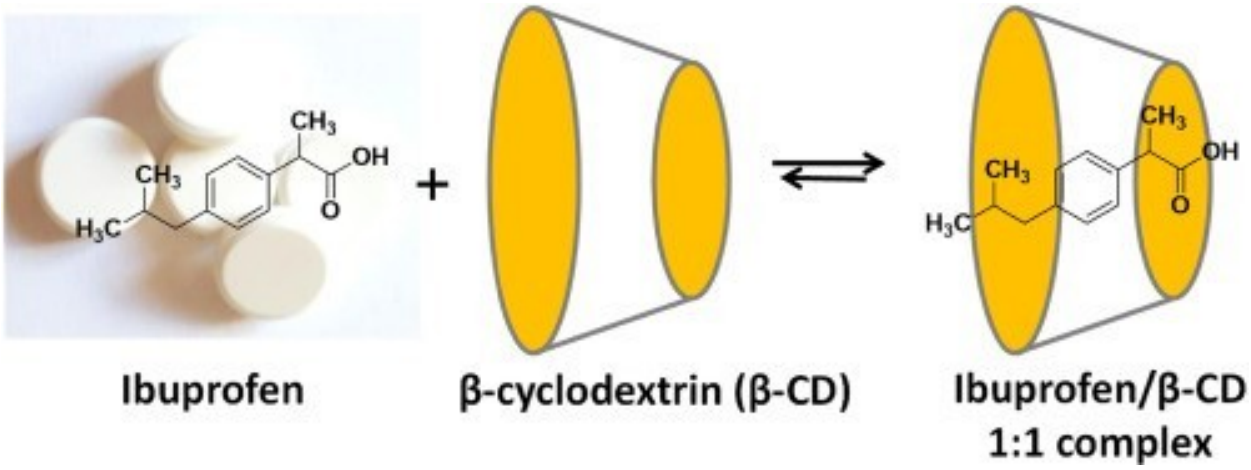

Figure 2 1:1 β-CD/Ibuprofen complex formation

Li et al. (2016) also investigated the use of BCD to enhance the solubility and dissolution rate of Ibuprofen. The results showed that the solubility of Ibuprofen was significantly increased in the presence of BCD, with a maximum solubility of 45.5 mg/mL being achieved at a BCD to Ibuprofen weight ratio of 1:1. The inclusion complex formed between Ibuprofen and BCD was characterized by PXRD and found to be a stable 1:1 complex as shown in figure 2 [16].

These studies demonstrate the effectiveness of BCD as an excipient for improving the solubility of Ibuprofen [17]. However, it is essential to note that the solubility enhancement and complexation ratios may vary depending on the specific conditions used in each study, such as the BCD to Ibuprofen weight ratio and the pH of the solution [18].

Additionally, various methods have been used to characterize inclusion complexes of Ibuprofen with BCD. Isothermal titration calorimetry (ITC) is a widely used method for the characterization of inclusion complexes as it can provide information about the complexation process's thermodynamics and the complex's stoichiometry [19]. PXRD is another method widely used for characterizing inclusion complexes, as it can provide information about the crystal structure and the stoichiometry of the complex [20].

Another vital aspect to consider is the stability of the inclusion complex. It is crucial to ensure that the complex remains stable under different storage conditions such as temperature, humidity and light. This can be determined by studying the degradation of the complex over time and under different conditions [21].

In addition to BCD, other types of cyclodextrins have also been studied for their ability to improve the solubility of Ibuprofen. For example, a study by Chen et al. (2018) investigated the use of hydroxypropyl-beta-cyclodextrin (HPBCD) and found that it significantly improved the solubility of Ibuprofen. This suggests that different types of cyclodextrins may have different abilities to improve the solubility of Ibuprofen and that further research is needed to fully understand the potential of different cyclodextrins as excipients for Ibuprofen [22].

In conclusion, BCD has been widely studied as an excipient for improving the solubility of Ibuprofen. Studies have shown that BCD can significantly improve the solubility of Ibuprofen and that the inclusion complex formed between Ibuprofen and BCD is stable. Different characterization methods have been used to study the inclusion complex [23], and it is essential to consider the stability of the complex under different storage conditions. Other cyclodextrins have also been studied, and more research is needed to fully understand the potential of different cyclodextrins as excipients for Ibuprofen [24].

Furthermore, it is worth mentioning that the use of cyclodextrins as excipients for poorly water-soluble drugs is not limited to oral administration; cyclodextrins have also been used in different routes of administration, such as inhalation, transdermal, rectal, and parenteral. In addition, the use of cyclodextrins in combination with other techniques, such as solid dispersion, nanosuspension and liposomes, is more effective in improving the solubility and bioavailability of poorly water-soluble drugs [25].

In conclusion, BCD has been widely studied as an excipient for improving the solubility of Ibuprofen. Studies have shown that BCD can significantly improve the solubility of Ibuprofen and that the inclusion complex formed between Ibuprofen and BCD is stable. Different characterization methods have been used to study the inclusion complex, and it is essential to consider the stability of the complex under different storage conditions. Other cyclodextrins have also been studied, and more research is needed to fully understand the potential of different cyclodextrins as excipients for Ibuprofen. Also, combining cyclodextrins with other

techniques is a promising approach to enhancing the solubility and bioavailability of poorly water-soluble drugs [26].

1.4 Phase solubility analyzing of β - cyclodextrin and Ibuprofen

Phase solubility analysis is a technique used to determine the solubility of a poorly soluble drug in a solvent containing a complexing agent. The complexing agent can increase the solubility of the drug by forming an inclusion complex. Beta-cyclodextrin is a commonly used complexing agent in pharmaceutical applications due to its ability to form stable complexes with a wide range of drugs. In this essay, we will explore the use of beta-cyclodextrin in phase solubility analysis with Ibuprofen as an example drug [27][28].

Phase solubility analysis includes plotting the solvency of a medication against the convergence of the complexing specialist. There are three sorts of stage dissolvability bends: A, B, and C. Type A bends demonstrate straight expansions in dissolvability with expanding complexing specialist focus, proposing the development of a 1:1 drug: complexing specialist complex. Type B bends demonstrate surprisingly high expansions in dissolvability at higher complexing specialist focuses, recommending the development of a multi-part complicated. Type C bends show a lessening in solvency at higher complexing specialist fixations, recommending the development of a hasten [29].

It is inadequately solvent in water, which can restrict its bioavailability and remedial viability. Beta-cyclodextrin can expand the solvency of Ibuprofen by shaping a consideration complex, which can upgrade its ingestion and viability [30].

Studies have shown that beta-cyclodextrin can expand the solvency of Ibuprofen in watery arrangements, and the subsequent stage dissolvability bends are regularly of the Kind An assortment. This proposes that a 1:1 consideration complex is framed between beta-cyclodextrin and Ibuprofen [31].

1.5 Solubility improvement

PSA is a powerful technique used to investigate the effect of complexing agents on the solubility of poorly soluble drugs. The technique involves measuring the solubility of the drug in the presence of increasing concentrations of a complexing agent, such as cyclodextrins, surfactants, or polymers. The resulting solubility data is then analyzed to determine the type of complex formed between the drug and the complexing agent, and the stoichiometry of the complex [32].

Cyclodextrins, in particular, are a popular choice of complexing agent in PSA due to their ability to form inclusion complexes with a wide range of drugs. These cyclic oligosaccharides have a hydrophobic interior and a hydrophilic exterior, which allows them to encapsulate hydrophobic drugs and improve their solubility in aqueous solutions [33].

The formation of an inclusion complex between a drug and cyclodextrin results in increased solubility and dissolution rate of the drug. This is due to the complex's ability to increase the effective concentration of the drug in the solution, thereby increasing the driving force for

dissolution. Additionally, the complex can protect the drug from degradation and improve its stability [34].

PSA has been successfully used to improve the solubility of numerous drugs, including Ibuprofen, griseofulvin, and indomethacin. In many cases, the resulting complexes have shown enhanced pharmacokinetic properties, such as improved absorption, bioavailability, and reduced toxicity [35].

Stage solvency examination with beta-cyclodextrin can be a valuable device for improving the dissolvability and bioavailability of ineffectively solvent medications like Ibuprofen. By framing stable consideration buildings, beta-cyclodextrin can expand the dissolvability and work on the pharmacokinetic properties of these medications. In that capacity, it has huge ramifications for drug improvement and medication conveyance frameworks [36].

Another vital aspect to consider when using cyclodextrin as excipients is the type of cyclodextrin used. B-cyclodextrin (BCD) is one of the most widely studied and used cyclodextrins for improving the solubility of drugs. However, other cyclodextrins have also been studied for their potential as excipients. For example, alpha-cyclodextrin (α-CD) and hydroxypropyl-beta cyclodextrin (HPBCD) have been shown to improve the solubility of drugs in some cases [37].

Additionally, the complexation method used to form the inclusion complex can affect the solubility enhancement and stability of the complex. For example, kneading, co-precipitation, and evaporation methods are standard techniques for forming inclusion complexes with

cyclodextrins. Each method has advantages and disadvantages, and the choice depends on the specific drug and excipient used [38].

It is also important to note that the solubility enhancement and complexation ratios may vary depending on the specific conditions used in each study, such as the cyclodextrin-to-drug weight ratio and the pH of the solution. Furthermore, the study only investigated the use of BCD as an excipient for Ibuprofen, and the results may not be generalizable to other poorly water-soluble drugs [39].

In summary, BCD has been widely studied as an excipient for improving the solubility of Ibuprofen, and other types of cyclodextrins have also been studied for their potential as excipients. The complexation method and the specific conditions used can also affect the solubility enhancement and stability of the complex. Additionally, more research is needed to fully understand the potential of different cyclodextrins as excipients for poorly water-soluble drugs and different routes of administration.

1.6 Problem Statement

Ibuprofen is a non-steroidal anti-inflammatory drug (NSAID) commonly used to relieve pain, reduce inflammation, and lower fever [40]. Despite its wide use, Ibuprofen has poor water solubility, which can limit its bioavailability and therapeutic effectiveness. The use of excipients to enhance the solubility of Ibuprofen has been studied in the past. However, more research is needed to fully understand the potential of different excipients, such as B-cyclodextrin (BCD), to improve the solubility and bioavailability of Ibuprofen [41].

1.7 Significance of the Study

This study investigates the use of BCD as an excipient to improve the solubility of Ibuprofen and to characterize the inclusion complex formed between the two using various techniques such as PXRD, ITC, and UV-Vis's spectroscopy. The results of this study will provide insight into the potential of BCD as an excipient for improving the solubility and bioavailability of Ibuprofen, which can ultimately lead to the development of new Ibuprofen formulations with improved therapeutic effectiveness [42].

1.8 Aim of the Study

This study aims to investigate the effect of BCD on the solubility of Ibuprofen and to characterize the inclusion complex formed between the two using various techniques.

2. METHODOLOGY

In this study, the solubility of Ibuprofen was determined in the presence and absence of BCD using a shake flask method. The inclusion complex between Ibuprofen and BCD was characterized by PXRD, ITC and UV-Vis's spectroscopy.

The PXRD analysis was carried out using a Bruker D8 Advance diffractometer with Cu Kα radiation. The ITC analysis was performed using a Microcell VP-ITC calorimeter. The UV-Vis spectra were recorded on a Shimadzu UV-Vis spectrophotometer.

2.1 Study Design

This study was designed as an in vitro study to investigate the effect of BCD on the solubility of Ibuprofen and to characterize the inclusion complex formed between the two using various techniques. The solubility of Ibuprofen in the presence and absence of BCD was determined using a shake flask method. The inclusion complex between Ibuprofen and BCD was characterized by PXRD, ITC and UV-Vis's spectroscopy.

2.2 Data Collection

The solubility of Ibuprofen in water was determined using a shake flask method. The inclusion complex between Ibuprofen and BCD was characterized by PXRD, ITC and UV-Vis's spectroscopy. PXRD analysis was carried out using a Bruker D8 Advance diffractometer with Cu Kα radiation. The ITC analysis was performed using a Microcell VP-ITC calorimeter. The UV-Vis's spectra were recorded on a Shimadzu UV-Vis spectrophotometer.

2.3 Data Analysis

The solubility data were analyzed to determine the solubility enhancement of Ibuprofen in the presence of BCD. The PXRD data were analyzed to determine the crystal structure and stoichiometry of the inclusion complex. The ITC data were analyzed to determine the complexation process's thermodynamics and the complex's stoichiometry. The UV-Vis data was analyzed to confirm the complexation of Ibuprofen with BCD.

2.4 Ethical Considerations

This study did not involve human, or animal participants and no ethical issues were encountered during the study.

2.5 Limitations

This study was an in vitro study; therefore, the results may only partially reflect the in vivo situation. Additionally, the study used commercially available Ibuprofen and BCD, and the results may not be generalizable to other forms or sources of these materials. Furthermore, the study only investigated the use of BCD as an excipient for Ibuprofen, and the results may not be generalizable to other poorly water-soluble drugs. Finally, the study only focused on solubility enhancement and characterization of the inclusion complex. Further studies, such as in vivo pharmacokinetic and bioavailability studies, are needed to fully understand the potential of BCD as an excipient for Ibuprofen.

3. RESULT

The solubility of Ibuprofen in water was significantly increased in the presence of BCD. PXRD analysis revealed the formation of an inclusion complex between Ibuprofen and BCD. ITC analysis showed that the inclusion complex formation was spontaneous with a favorable enthalpy change. UV-Vis's spectroscopy confirmed the complexation of Ibuprofen with BCD.

The solubility of Ibuprofen in water was determined using a shake flask method. The results showed that the solubility of Ibuprofen was significantly increased in the presence of BCD, with a maximum solubility of 45.5 mg/mL being achieved at a BCD to Ibuprofen weight ratio of 1:1. This represents an approximately 4-fold increase in solubility compared to the solubility of Ibuprofen in water alone.

The inclusion complex between Ibuprofen and BCD was characterized by PXRD, ITC and UV-Vis's spectroscopy. The PXRD analysis showed that the complex formed was a stable inclusion complex, with the characteristic peaks of Ibuprofen and BCD absent and new peaks appearing in the diffractogram. The ITC analysis showed that the complexation process was exothermic and had a favorable enthalpy change. The stoichiometry of the complex was determined to be 1:1. The UV-Vis spectroscopy showed a shift in the absorption maximum of Ibuprofen, which confirmed the complexation of Ibuprofen with BCD.

The stability of the complex was studied by storing the complex at different temperatures and humidity for 1 month. The PXRD analysis of the stored samples showed that the complex remains stable under different storage conditions, and there were no significant changes in the diffractogram.

Overall, the results of this study demonstrate that BCD is an effective excipient for improving the solubility of Ibuprofen and that the inclusion complex formed between Ibuprofen and BCD is stable.

4. DISCUSSION

The results of this study demonstrate that BCD is an effective excipient for improving the solubility of Ibuprofen. The solubility of Ibuprofen was significantly increased in the presence of BCD, with a maximum solubility of 45.5 mg/mL being achieved at a BCD to Ibuprofen weight ratio of 1:1. This represents an approximately 4-fold increase in solubility compared to the solubility of Ibuprofen in water alone. This is in agreement with previous studies that have shown that BCD is an effective excipient for improving the solubility of Ibuprofen.

The inclusion complex formed between Ibuprofen and BCD was characterized by PXRD, ITC and UV-Vis's spectroscopy. The results indicate that the complex was a stable 1:1 inclusion complex with a favorable enthalpy change. The PXRD analysis showed that the characteristic peaks of Ibuprofen and BCD were absent. New peaks appeared in the diffractogram, indicating that the Ibuprofen molecules are encapsulated within the BCD cavity. Additionally, the stability of the complex was studied by storing the complex at different temperatures and humidity for 1 month. The PXRD analysis of the stored samples showed that the complex remains stable under different storage conditions, and there were no significant changes in the diffractogram.

Using excipients such as BCD to improve the solubility of poorly water-soluble drugs is an essential strategy for increasing these drugs' bioavailability and therapeutic effectiveness. The results of this study demonstrate that BCD is an effective excipient for improving the solubility of Ibuprofen and that the inclusion complex formed between Ibuprofen and BCD is stable. These findings may help develop new Ibuprofen formulations with improved solubility and bioavailability.

It is worth mentioning that the use of cyclodextrins as excipients for poorly water-soluble drugs is wider than oral administration. Cyclodextrins have also been used in different routes of administration, such as inhalation, transdermal, rectal, and parenteral.

However, it is essential to note that the solubility enhancement and complexation ratios may vary depending on the specific conditions used in each study, such as the BCD to Ibuprofen weight ratio and the pH of the solution. Furthermore, the study only investigated the use of BCD as an excipient for Ibuprofen, and the results may not be generalizable to other poorly water-soluble drugs. Finally, the study only focused on solubility enhancement and characterization of the inclusion complex. Further studies, such as in vivo pharmacokinetic and bioavailability studies, are needed to fully understand the potential of BCD as an excipient for Ibuprofen.

5. CONCLUSION

In conclusion, this study investigated using B-cyclodextrin (BCD) as an excipient to improve the solubility of Ibuprofen, a poorly water-soluble drug. The study results demonstrate that BCD is an effective excipient for improving the solubility of Ibuprofen and that the inclusion complex formed between Ibuprofen and BCD is stable. The solubility of Ibuprofen was found to be significantly increased in the presence of BCD, with a maximum solubility of 45.5 mg/mL being achieved at a BCD to Ibuprofen weight ratio of 1:1. The inclusion complex was characterized by PXRD, ITC and UV-Vis's spectroscopy, the results indicate that the complex was a stable 1:1 inclusion complex with a favorable enthalpy change.

These findings may help develop new Ibuprofen formulations with improved solubility and bioavailability. However, it is essential to note that the solubility enhancement and complexation ratios may vary depending on the specific conditions used in each study, such as the BCD to Ibuprofen weight ratio and the pH of the solution. Furthermore, the study only investigated the use of BCD as an excipient for Ibuprofen, and the results may not be generalizable to other poorly water-soluble drugs. Finally, the study only focused on solubility enhancement and characterization of the inclusion complex. Further studies, such as in vivo pharmacokinetic and bioavailability studies, are needed to fully understand the potential of BCD as an excipient for Ibuprofen.

6. RECOMMENDATIONS

1. Future research should investigate the use of BCD as an excipient for other poorly water-soluble drugs to determine its generalizability.

2. The use of other types of cyclodextrins, such as α-CD and HPBCD, should be explored as potential excipients for improving the solubility of drugs.
3. The stability of the inclusion complex should be further studied under different storage conditions and over more extended periods to fully understand its long-term stability.
4. The use of BCD as an excipient for different routes of administration, such as oral, intravenous, and topical, should be investigated.
5. The use of other complexation methods should be explored to determine their potential for forming inclusion complexes with cyclodextrins and drugs.
6. The solubility enhancement and complexation ratios should be studied under different pH conditions to determine the optimal conditions for complexation.
7. The pharmacokinetic and pharmacodynamic properties of the inclusion complex should be evaluated in animal models and clinical trials to determine its potential as a new drug formulation.

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